Chinatip Lawansuk（廖有福）

Taipei, Taiwan | Email: l.chinatip@gmail.com | Mobile: +886987422475 | Linkedin: Chinatip Lawansuk
Github: github.com/chinatip-l

Professional Summary

Driven by a passion for building robust systems from low level, I began my career as an Embedded System Engineer, where I developed firmware and infotainment systems for a fleet of over 1000 electric vehicles. To expand my knowledge from the system level to the chip level, I pursued a Master's degree in Electrical Engineering, focusing on digital IC design. This advanced study allowed me to dive deep into hardware design and optimisation, culminating in a thesis where I designed and implemented a CORDIC hardware accelerator with a custom caching system.

Expertises

Programming: Verilog, SystemVerilog, C, C++, Python, Assembly, Shell Script, TCL, Tensorflow
IC Design Software: Synopsys VCS, Synopsys Design Compiler (Design Vision), Synopsys TetraMAX, Cadence NC-Verilog, Cadence Innovus, Siemens Calibre
Skills: Cell-Based Digital IC Design Flow, Image Processing, Computer Architecture, Operating System, Embedded System Design, Firmware, RTOS, Circuit Design
Coursework: Computer-Aided VLSI System Design, Mixed-Signal IC Design, Computer Vision, Computer Architecture, Computer Organisation, Operating System

Research and Publication

CORDIC hardware Improvement using Content Addressable Memory Cache (Thesis)

- Investigated and designed an algorithm to cache partially calculated data and reuse it in the CORDIC architecture for sine and cosine calculation.
- Designed and implemented the processing core to suit the caching algorithm
- Designed and implemented a custom TCAM to use with the proposed cache unit and a complete cache subsystem from scratch, including multiple cache blocks, a PLRUm replacement policy, and a controller
- Designed and implemented a pipeline and synchronisation mechanism using a handshaking protocol
- Completed the full cell-based IC design flow from logic design and verification, synthesis, Design For Testability, Automatic Place and Route, post-layout simulation, and transistor-level simulation
- Using TSMC 90nm technology, resulting in a peak throughput of 67.796 MOPS, consumes 48.2 mW @ 338.98 MHz.

Two-Dimensional Convolution Accelerator of High-Precision Quantization (Conference Paper)

- Y. -T. Chao, J. -Y. Gao, **C. Lawansuk** and Y. -C. Fan, "Two-Dimensional Convolution Accelerator of High-Precision Quantization," *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA, 2024, pp. 1-2, doi: 10.1109/ICCE59016.2024.10444315.
- Developed a high-precision quantized 2D convolution accelerator leveraging the Winograd algorithm, achieving <0.2% quantization error.

Education

National Taipei University of Technology (國立臺北科技大學) Sept 2023 –Aug 2025
MSc in Electrical Engineering (GPA: 4.00/4.00)
- **Laboratory:** Integrated System Laboratory (ISLAB) @ NTUT

King Mongkut's Institute of Technology Ladkrabang (KMITL) Aug 2017 –Jul 2021
BE in Computer Engineering (GPA: 3.29/4.00)

Awards

First-Ranked Graduating Student (National Taipei University of Technology) May 2025
Second Class Honour (KMITL) Jul 2021

Experience

Firmware Engineer Intern, LinkCom Manufacturing Co., Ltd. – New Taipei, Taiwan Aug 2024 - May 2025
- Worked on a high-power-density DC-DC Converter including hardware design (Planar Transformer, LLC topology) and embedded software optimisation. Performance reaches more than 1 kW on the credit card size module.
- Initiate the project for the transformer test platform (Firmware, and Hardware)
- Investigate and design closed loop control system for power module and identify impact parameters

Embedded System Engineer, MuvMi Co., Ltd. – Bangkok, Thailand Oct 2021 - Sept 2023
- Develop IoT Fleet, Data Analysis and Visualisation Platform on AWS for EV Charging System, identified and mitigated up to 12% of energy loss
- Develop a Vehicle Control Unit Firmware (RTOS on arm Cortex M4 processor) and infotainment system (Buildroot on arm Cortex A7 processor) for Electric Minibus and Electric Tuk-Tuk for the fleet of 1000 vehicles